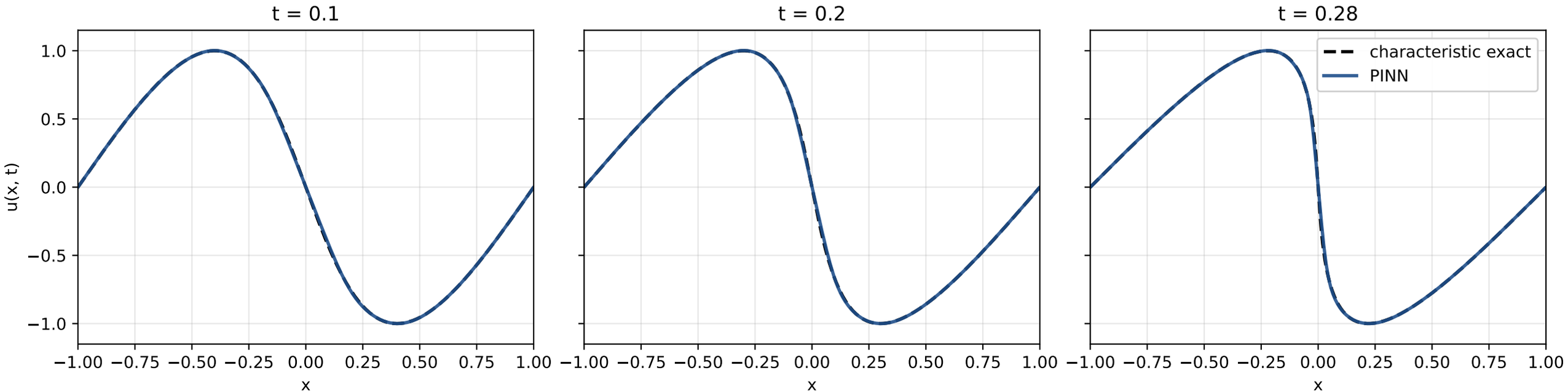
Inviscid Burgers PINN snapshots vs characteristic exact (training $t \in [0, 0.30]$, $t^*_{\text{break}} = 1/\pi \approx 0.318$)
t = 0.1
t = 0.2
t = 0.28
characteristic exact
PINN
u(x, t)
x
x
x
1.0
0.5
0.0
−0.5
−1.0
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00